Research Proposal

Machine-Learning Based Flash Flood Susceptibility Mapping Integrating Hajj-Season Population Density in the Wadi Mehassar Basin and Al Mashaaer Al Muqaddassa, Makkah, Saudi Arabia

1. Abstract

Flash floods pose a serious threat to the rapidly urbanising eastern part of Makkah and the holy sites of Al Mashaaer Al Muqaddassa, particularly during the Hajj season when millions of pilgrims congregate within the Wadi Mehassar basin. This study proposes a practical machine-learning framework, using Random Forest and XGBoost, to map flash-flood susceptibility across the basin. Ten conditioning factors including a purpose-built Hajj-season population density layer will be integrated within a GIS environment and used to train both classifiers. The resulting susceptibility surfaces will be validated against Sentinel-1-derived flood extents and cross-checked against previously published HEC-RAS hydraulic modelling results. A dedicated with/without-exposure-factor comparison will quantify how the inclusion of pilgrimage-season population density reshapes the spatial pattern of high-susceptibility zones. The work is scoped to be technically feasible at master's level while producing actionable spatial information for Makkah Municipality, Hajj authorities and civil defence agencies.

2. Introduction and Background

Makkah is located in an arid, mountainous environment characterised by steep slopes, ephemeral wadis and short-duration, high-intensity rainfall. The Wadi Mehassar basin (approximately 86 km²) drains toward the eastern urban districts and the pilgrimage zones of Mina, Muzdalifah and Arafat. Previous studies of this basin have relied mainly on hydrological–hydraulic models (HEC-1, HEC-RAS) to simulate inundation extents. While these models remain the standard for engineering design, they do not by themselves produce continuous, probabilistic susceptibility surfaces, nor do they explicitly incorporate the extreme, temporally concentrated population exposure that characterises the Hajj period.

Machine-learning methods, particularly Random Forest and XGBoost, have become a standard modern complement to hydraulic modelling for flood susceptibility mapping worldwide, because they handle non-linear relationships and mixed data types, and yield both a continuous probability surface and a ranked measure of variable importance. Applying these methods to Wadi Mehassar, with an explicit pilgrimage-exposure factor, addresses a clear and currently unfilled gap in the local literature.

3. Problem Statement and Research Gap

- Existing studies of Wadi Mehassar focus on hydraulic modelling and do not generate machine-learning-based susceptibility maps.
- Hajj-season population density has rarely been used as an explicit conditioning factor in susceptibility models for the holy sites, and no established method for spatialising it at the basin scale is currently in use locally.
- There is a shortage of simple, reproducible, decision-oriented susceptibility products that local authorities can update themselves and use directly for land-use planning and emergency response.

This research addresses these gaps by developing a reproducible machine-learning susceptibility model that explicitly incorporates pilgrimage exposure, and by testing rather than assuming how much that factor changes the resulting risk picture.

4. Research Objectives

1. Delineate the Wadi Mehassar basin and prepare a consistent set of conditioning-factor rasters at a common resolution.
2. Develop Random Forest and XGBoost models to generate a continuous flash-flood susceptibility surface.
3. Construct and integrate a Hajj-season population density layer as a conditioning factor, using a documented spatial disaggregation method rather than uniform basin-wide values.
4. Quantify the specific effect of the Hajj population density factor by comparing model outputs with and without it included.
5. Validate the susceptibility map against Sentinel-1 flood extents and compare it with published HEC-RAS results.
6. Produce practical maps and recommendations for urban planning and Hajj risk management.

Research Questions

- How does the inclusion of Hajj-season population density change the spatial pattern and extent of high-susceptibility zones, relative to a model built on physical factors alone?
- Which physical and socio-economic factors most strongly influence flash-flood susceptibility in Wadi Mehassar according to Random Forest and XGBoost variable-importance rankings, and how consistent are the two models' rankings?
- How well do the machine-learning susceptibility zones agree, spatially, with existing HEC-RAS-derived hazard maps?

5. Study Area

The study area is the Wadi Mehassar drainage basin (≈86 km²), located east of Makkah city, Kingdom of Saudi Arabia (approximately 39°50′–39°56′ E and 21°18′–21°27′ N). Elevations range from about 218 m a.m.s.l. at the outlet to more than 900 m a.m.s.l. in the upstream mountains. The basin includes parts of Al Mashaaer Al Muqaddassa (Mina, Muzdalifah and Arafat) and adjacent urban districts. The climate is arid, drainage is ephemeral, and the basin experiences both rapid permanent urban expansion and an extreme, short-duration seasonal population surge during Hajj a combination that makes it a particularly demanding but instructive test case for exposure-aware susceptibility mapping.

6. Methodology

6.1 Data Collection

- Digital Elevation Model: SRTM 30 m (free).
- Satellite imagery: Sentinel-1 (SAR) and Sentinel-2 (optical) free via Copernicus.
- Rainfall: published intensity values or National Center for Meteorology (NCM) station data.
- Soil / geology: existing published maps.

- Population: national census data and published Hajj capacity/occupancy figures for Mina, Muzdalifah and Arafat.
- Reference flood information: published HEC-RAS inundation maps and one or two Sentinel-1-observed flood events (e.g., April 2021).

6.2 Conditioning Factors

- Elevation
- Slope
- Topographic Wetness Index (TWI)
- Drainage density
- Distance to main wadi channels
- Land use / land cover
- NDVI
- Hajj-season population density

All factors will be derived in ArcGIS Pro or QGIS and resampled to a common 30 m grid. Because slope, TWI, drainage density and distance-to-channel are physically related, pairwise correlation and Variance Inflation Factor (VIF) diagnostics will be run on the factor set before model training; any factor pair with VIF above an accepted threshold (commonly 5–10) will be flagged and, if needed, one member of the pair dropped or the factors combined, so that the variable-importance ranking reported in the results is not distorted by redundancy between predictors.

6.3 Hajj-Season Population Density Layer

Rather than applying a single basin-wide population figure, Hajj-season population will be spatialised using a dasymetric approach: published pilgrim capacity and occupancy figures for defined zones (Mina tent sectors, Muzdalifah encampment areas, Arafat plain) will be redistributed within each zone's built or usable footprint derived from Sentinel-2 land cover and known camp/tent-block boundaries rather than applied uniformly across each zone's full administrative area. This produces a gridded density surface at the same 30 m resolution as the other factors, with the redistribution assumptions (which zones, which capacity source, which footprint mask) documented explicitly so the layer can be reproduced or updated by local authorities with newer occupancy data in future Hajj seasons.

6.4 Flood Inventory

A binary flood/non-flood raster will be created from available Sentinel-1 imagery for one or two known flood events, using thresholding or visual interpretation. Given the limited number of source events, sample points will additionally be checked for spatial autocorrelation, and a spatial (rather than purely random) train/test split will be used withholding a contiguous sub-region of the basin for testing so that reported accuracy is not inflated by neighbouring flood/non-flood points leaking between the training and test sets.

6.5 Machine-Learning Modelling

7. Export factor values and flood labels as a table of sample points (balanced flood and non-flood samples).
8. Split data into training (70%) and testing (30%) sets using a spatial split, as noted in 6.4.
9. Train Random Forest and XGBoost models (Python, scikit-learn) on the full ten-factor set.

10. Train a second pair of models with the Hajj-season population density factor excluded, to serve as the physical-factors-only baseline for Objective 4.
11. Predict flood probability across the basin with both factor sets to produce continuous susceptibility surfaces.
12. Reclassify the probability maps into five susceptibility classes (Very Low to Very High) using Natural Breaks or quantiles.
13. Rank variable importance from both models and compare rankings for consistency.

6.6 With/Without Population-Factor Comparison

The with- and without-Hajj-factor susceptibility surfaces (6.5, step 4) will be differenced and cross-tabulated by susceptibility class to identify where and by how much the classification shifts. This provides a direct, quantitative answer to Research Question 1, rather than relying on visual comparison alone.

6.7 Validation

- Calculate ROC–AUC, accuracy, precision, recall and F1-score on the spatially held-out test set, for both the full and physical-factors-only models.
- Visually and quantitatively compare high-susceptibility zones against published HEC-RAS inundation maps.
- Report the class-transition results from Section 6.6 as the formal measure of the Hajj population factor's spatial effect.

6.8 Software

- GIS: ArcGIS Pro or QGIS.
- Machine learning: Python (Google Colab or local Jupyter) with scikit-learn and XGBoost.
- No specialised commercial hydraulic software is required.

7. Expected Outcomes

- A five-class flash-flood susceptibility map of Wadi Mehassar, produced with and without the Hajj population density factor.
- A documented, reproducible method for spatialising Hajj-season population density as a GIS raster layer.
- A quantitative measure of how much and where the pilgrimage-exposure factor changes the susceptibility classification.
- Variable-importance rankings from both Random Forest and XGBoost, with a multicollinearity check reported alongside them.
- Comparison maps against previously published HEC-RAS hydraulic results.
- Practical recommendations for land-use zoning and emergency planning in the holy sites.
- Complete Python scripts and GIS workflows that local authorities can reuse or update with newer data.

8. Significance

The study will deliver the first machine-learning susceptibility map for the Wadi Mehassar basin that explicitly incorporates, and quantifies the effect of, Hajj-season population density. Because the exposure layer's construction method and the with/without comparison are both fully documented, the results are intended to be directly usable and updatable by Makkah Municipality and Hajj authorities, supporting safer urban planning, better-targeted drainage infrastructure investment, and improved crowd-management strategies during the pilgrimage season.

10. Key References (Selected)

El-Saoud, W. A., & Othman, A. (2022). An integrated hydrological and hydraulic modelling approach for flash flood hazard assessment in eastern Makkah city, Saudi Arabia. Journal of King Saud University – Science.

Recent machine-learning flood susceptibility studies using Random Forest and XGBoost in arid environments (to be expanded during the literature review).